

# Background

This tool has six categories of performance, and each are weighted when calculating an overall ‘utility’ for the package. It is not currently possible to change the weightings – that will require time we don’t have right now 😊 .

Table 1 - Category weighting

| Performance Category                  | Default weighting (percent) |
|---------------------------------------|-----------------------------|
| Crop value                            | 50                          |
| Direct costs                          | 12.5                        |
| Environmental impact                  | 12.5                        |
| Human health and safety               | 12.5                        |
| Time and management                   | 6.25                        |
| Third party coordination requirements | 6.25                        |

Table 2 – User provided ratings

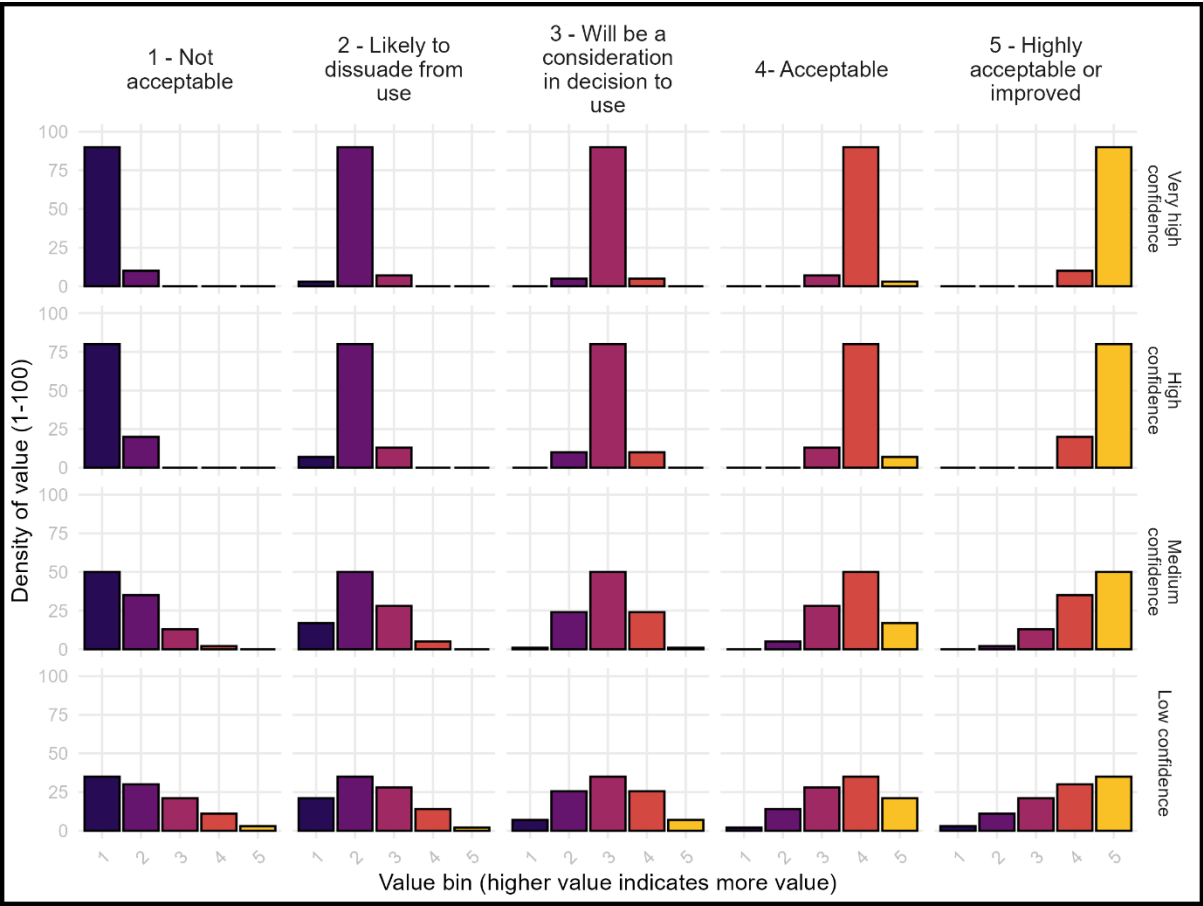
| User Rating                                    | Assigned Value |
|------------------------------------------------|----------------|
| 1 – Not acceptable                             | 1              |
| 2 – Likely to dissuade from use                | 2              |
| 3 – Will be a consideration in decision to use | 3              |
| 4 - Acceptable                                 | 4              |
| 5 – Highly acceptable or improved              | 5              |

Note that you will need to assign a confidence to your assessments (Table 3).

Table 3 – User provided confidences

| Confidence rating | Percent of distribution allotted to selected rating |
|-------------------|-----------------------------------------------------|
| Very high         | 90%                                                 |
| High              | 80%                                                 |
| Medium            | 50%                                                 |
| Low               | 35%                                                 |

Figure 1 shows how the user rating and confidence scores are combined to create the ‘value distribution’.



# Questionnaire

The following questionnaire should be filled out for both Package #1 and #2.

## Crop value

This is a measure of the success and therefore the acceptability of the yields and quality obtained. The higher the yield and/or quality the greater the acceptability.

Loss in crop yield and/or quality when using the technology/package is:

|                    |                                 |                                                |                |                                   |
|--------------------|---------------------------------|------------------------------------------------|----------------|-----------------------------------|
| 1 – Not acceptable | 2 – Likely to dissuade from use | 3 – Will be a consideration in decision to use | 4 - Acceptable | 5 – Highly acceptable or improved |
|--------------------|---------------------------------|------------------------------------------------|----------------|-----------------------------------|

My confidence in this rating is:

|     |        |      |           |
|-----|--------|------|-----------|
| Low | Medium | High | Very high |
|-----|--------|------|-----------|

Some things to consider:

- Opportunity costs when implementing the practice(s), for example if you take strips of land out of production to install natural habitat to support natural enemies.
- An increase in crop value due to new markets available, for example by producing pesticide-free products

## Direct costs

This is a measure of the value delivered against the cost of the package.

The net cost of the technology/package to implement is:

|                    |                                 |                                                |                |                                   |
|--------------------|---------------------------------|------------------------------------------------|----------------|-----------------------------------|
| 1 – Not acceptable | 2 – Likely to dissuade from use | 3 – Will be a consideration in decision to use | 4 - Acceptable | 5 – Highly acceptable or improved |
|--------------------|---------------------------------|------------------------------------------------|----------------|-----------------------------------|

My confidence in this rating is:

|     |        |      |           |
|-----|--------|------|-----------|
| Low | Medium | High | Very high |
|-----|--------|------|-----------|

Some things to consider:

- Does it require upfront costs that take time to see a return on investment, for example buying a mechanical weeding implement.
- Does it require time before the return on investment is realized, or is it realized immediately, for example planting of a diverse flower buffer versus re-establishing native habitat.

## Environmental impact

Please consider both the pesticides and non-pesticide based impacts in this assessment. The environmental costs from pesticides can be calculated from the pesticide application data using the Pesticide Impact Explorer (PIE) tool: [agro.au.dk/pietool](http://agro.au.dk/pietool)

. The potential environmental impact of the practice(s) is/are:

|                    |                                 |                                                |                |                                   |
|--------------------|---------------------------------|------------------------------------------------|----------------|-----------------------------------|
| 1 – Not acceptable | 2 – Likely to dissuade from use | 3 – Will be a consideration in decision to use | 4 - Acceptable | 5 – Highly acceptable or improved |
|--------------------|---------------------------------|------------------------------------------------|----------------|-----------------------------------|

My confidence in this rating is:

|     |        |      |           |
|-----|--------|------|-----------|
| Low | Medium | High | Very high |
|-----|--------|------|-----------|

Some things to consider:

- Does the practice impact greenhouse gas emissions, for example by requiring more tractor passes
- Does the practice impact water quality, for example by reducing nitrogen applications
- Does the practice support biodiversity, for example by establishing flower strips along field borders

## Human health and safety

Please consider both the pesticides and non-pesticide based impacts in this assessment. The human health costs from pesticides can be calculated from the pesticide application data using the Pesticide Impact Explorer (PIE) tool: [agro.au.dk/pietool](http://agro.au.dk/pietool)

Human health and safety risks are:

|                    |                                 |                                                |                |                                   |
|--------------------|---------------------------------|------------------------------------------------|----------------|-----------------------------------|
| 1 – Not acceptable | 2 – Likely to dissuade from use | 3 – Will be a consideration in decision to use | 4 - Acceptable | 5 – Highly acceptable or improved |
|--------------------|---------------------------------|------------------------------------------------|----------------|-----------------------------------|

My confidence in this rating is:

|     |        |      |           |
|-----|--------|------|-----------|
| Low | Medium | High | Very high |
|-----|--------|------|-----------|

Some things to consider:

- Exposure to physical risks, such as the use of ladders when implementing the practice
- Does it involve activities that must be done at night, or during the heat of the day
- You should think about health and safety issues not relating to the use of pesticides – there is an additional tool (Pesticide Impact Explorer (PIE) that allows you to enter the pesticide applications and get a numeric estimate of risk, as well as societal costs

## Time and management

The impact of implementing the package on time and management considerations, outside of those captured by direct costs.

*The time and/or effort requirement of the user to implement and/or maintain the effectiveness of the practice(s) is:*

|                    |                                 |                                                |                |                                   |
|--------------------|---------------------------------|------------------------------------------------|----------------|-----------------------------------|
| 1 – Not acceptable | 2 – Likely to dissuade from use | 3 – Will be a consideration in decision to use | 4 - Acceptable | 5 – Highly acceptable or improved |
|--------------------|---------------------------------|------------------------------------------------|----------------|-----------------------------------|

My confidence in this rating is:

|     |        |      |           |
|-----|--------|------|-----------|
| Low | Medium | High | Very high |
|-----|--------|------|-----------|

Some things to consider:

- Does the practice require certain skills or training?
- Does the practice require careful timing of application?

## Third party coordination requirements

Whether implementation requires additional supply chains, infrastructure, or cooperation that may be outside of the user's direct control.

*The coordination required for the technology/package to be effective is:*

|                    |                                 |                                                |                |                                   |
|--------------------|---------------------------------|------------------------------------------------|----------------|-----------------------------------|
| 1 – Not acceptable | 2 – Likely to dissuade from use | 3 – Will be a consideration in decision to use | 4 - Acceptable | 5 – Highly acceptable or improved |
|--------------------|---------------------------------|------------------------------------------------|----------------|-----------------------------------|

My confidence in this rating is:

|     |        |      |           |
|-----|--------|------|-----------|
| Low | Medium | High | Very high |
|-----|--------|------|-----------|

Some things to consider:

- Can a producer implement the practice(s) immediately, or do they require training to run a spot sprayer
- Can the producer implement it now, or do they need to hire a third-party to, for example, spray pesticides using a drone
- Does the practice require establishment of supply chains, for example a new biological control agent or a new crop variety
- Does the practice require coordination with neighbors, for example establishment of buffer areas